AADIT AGRAWAL

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EDUCATION

Manipal Institute of Technology, MAHE — Jul 2023 – May 2027
BTech — Computer and Communication Engineering

EXPERIENCE

Head Student Researcher · *CASIR, MAHE* · *ecitis.org* — May 2025 – Present

- Architected **HPC cluster** (96-core Threadripper 7995WX, 48GB A6000, 256GB RAM) — **1,500+ daily inference req**, **99.2% uptime**.
- Leading the setup of a **$200K+ Compute Cluster using Apple Silicon + NVIDIA Blackwell**; optimising for ARM64 + CUDA.
- Real-time CV on **Jetson + DJI drone** — **30fps** edge inference via RTMP.
- **LLM semantic translator** for **Schneider Electric** — IEC 61131-3 → IEC 61499 conversion automation for legacy PLC codebases; migration effort reduced by **70%**.
- **Technical document ingestion pipeline** (Schneider Electric) for machine / electronics engineering support — evolved from RAG + BM25 + keyword cluster indexing to trigram indices with custom pi.dev-based search agents. VLM + LLM stack across **2,000+ docs**, **94% accuracy**, **3× faster** than the prior solution.
- **Application security analysis platform** (Schneider Electric) — source + compiled-binary evaluation in computer-use sandboxes with OS state monitoring (registry / DLL deltas), pattern-matched against a custom-correlated **CVE / CWE / MITRE** index. Pentest lead time reduced from **3 weeks → 3 days**.
- **3 production AI systems**; **2 peer-reviewed publications** in progress; team learnings published at blog.ecitis.org.

DevOps Engineer, Intern · *Pixa AI, Bengaluru* — Dec 2025

- Provisioned GPU compute (H100/H200/B200) and CI/CD pipelines for TTS/STT model fine-tuning on Phi-4; achieved **99.5% deployment success** across 2 API releases.

Frontend Engineer · *Rossmann New Media 2 Inc.* — Jan 2026 – Present

- Migrated rossmanngroup.com from WordPress to Next.js; improved page load performance and cross-platform responsiveness.
- Implemented technical SEO overhaul targeting data recovery keywords — moved primary rankings from 100+ to top 5.

Founding President · *PRISM* — Jan 2026 – Present

- Research club (AI/ML, SWE, HPC, DevOps, Biotech, InfoSec) — **150+ members**. Academic Advisory Board for AI Curriculum and Adoption.

Management Committee · *MIST* — Oct 2023 – Jun 2025

- CTF infra for **300+ users**, **100% uptime**; cut costs by **80%**.
- **12+ workshops** for **150+ participants**; top 10% national CTFs.

RESEARCH

FL + Digital Twin for Route Optimisation · *SCE, MAHE* — Jun 2024 – Jul 2025

- GPS telemetry (**500K+ datapoints**); dashcam CV via OpenCV + YOLO — **12,000+ labelled samples**.
- Built FL pipeline for privacy-preserving route optimization across distributed edge nodes.

PROJECTS

Homelab Infrastructure · *Self-hosted cloud* — 2022 – Present

- **70+ services** on Proxmox/Docker — **98% uptime**, **4 yrs**, 8TB MergerFS, Blackwell-GPU, 3-2-1 backups.
- Deployed custom object-detection models on security camera feeds achieving **30fps** real-time inference; offloaded **95%** of compute from CPU to GPU including 4K media transcoding.
- Mitigated **live DDoS** (**50K+ req/min**); **95%** attack surface reduction.

The TIK Menu API · GitHub — 2025 – Present

- **5,000+ DAU**, **150K+ monthly req**, campus adoption in under 7 days. REST API + Cloudflare R2.

Coolstuff.work Forgejo — 2023 – Present

- Suite of utility tools solving everyday problems for peers — **10,000+ aggregate DAU** across sub-apps.
- Built **OR-Free**, a real-time AI model availability and pricing tracker for OpenRouter, and **MIT Courses Search**, a course discovery tool with interactive graph visualization (**500+ nodes**). Next.js, Radix UI.

- Additional tools: transcript-cli (video transcription), Campus Directory (crowdsourced contacts), Mass Contact Maker (bulk VCF generation). See git.aadit.cc.

ophoner · *Mobile agent control loop* 2026 – Present

- Phone-based agent loop for device APIs and filesystem workflows — extensible via **tmux** and Android shell tooling. Focused on on-the-go file editing/querying without heavy desktop dependencies.

Developer Tooling (OSS) 2025 – Present

- agentskills: Published library of Claude Code skills for reusable agentic development workflows.
- t3code (fork): customised for parallelised agent-driven development flows.
- Self-hosted CI + sandboxing stack (migrating from Daytona); CodeRabbit integrated for public-repo review.

SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Go, Zig, Rust, Bash, SQL
Infrastructure & DevOps: Docker, Proxmox, Linux, Git, GitHub Actions, Terraform, Ansible, Nginx, Caddy, Tailscale
Cloud & Platforms: AWS, GCP, Azure, Cloudflare Workers, Vercel
AI/ML & HPC: PyTorch, JAX, Transformers, CUDA, SGLang, vLLM, llama.cpp, TensorRT-LLM, OpenCV, YOLO, NCCL
Web: Next.js, React, Svelte, Astro, Tailwind, Node.js, Bun, FastAPI, tRPC
Databases: PostgreSQL, MongoDB, SQLite, Redis, pgvector, ChromaDB
AI Tooling: Pi (Custom Implementation), Codex, Claude Code, Droid, Cursor, opencode, ampcode, CodeRabbit, MCP, Skills